

C S DEEPAK

Agentic AI / Applied AI Engineer

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deepak-ai-web.vercel.app — ask Dex, an AI assistant grounded in my work

EDUCATION

PES University, RR Campus

B.Tech in Computer Science Engineering

2023 – 2027

Bangalore, India

EXPERIENCE

Center for Data Sciences & Applied ML (CDSAML / CCBD)

Jun 2026 – Present

Agentic AI Researcher & Engineer – PES University

Bangalore, India

- Research on long-term memory and routing for autonomous multi-agent LLM systems; lead a 4-person team and own the memory schema and storage module.
- Designed evaluation protocols measuring owner-selection accuracy, handoff correctness and subtask assignment across multi-seed, multi-regime configurations.
- Drove ownership-scoring and fallback-routing decisions that cut token cost **23.84%** at equal answer accuracy with **0 retrains** across 4 drift regimes.

SahayAI Club, PES University

2025 – 2026

Head of Events and Management

Bangalore, India

- Ran **VICHAR 2026**, an AI ideathon with **100+ participants** across multiple structured problem-statement rounds.
- Shipped a 1,300-line frontend contribution to the club's live Next.js site (content-as-code pipeline, full SEO, handover docs for non-technical successors).

Nexus PES

Oct 2024 – Dec 2025

App Developer (Part-time)

Bangalore, India

- Android application development in Java.

PROJECTS

HandCode — Effect-Safety Control Plane for LLM Agents

Python, LiteLLM, SQLite

- Guarantees a crashed long-running agent never repeats an irreversible effect: every effect is classified and written to a fenced write-ahead ledger before it executes.
- **502 tests** including a nine-point chaos suite that kills the real process at each point; four end-to-end crash experiments with **zero duplicate effects**.
- CI reproduces the correctness claim on Linux and Windows, so it holds on a machine that is not mine.

ASMOS — Adaptive Semantic Memory for Multi-Agent LLMs

Python, Agentic AI

- Agents share verified knowledge checkpoints rather than chat history; a memory manager retains by utility and lets temporary detail expire.
- **23.84% fewer LLM tokens per query** at equal answer accuracy versus a global-memory baseline, causally tested and stamped to a named artifact.
- Ablation isolates the cause: freezing ownership evolution costs **+39.14 tokens/query** and never inverts across seeds. One hypothesis returned a null and is reported as such.

Warden — Trust Layer for Agentic Payments

Razorpay AI Buildathon 2026

- Safety layer between an AI agent and real money, catching both refund redirection and the quieter failure – suppressing a legitimate refund, which looks like a resolved case.
- **131 tests** over a 38-attack / 15-benign set; **14 models across 6 labs** measured against the same battery; exercised on a live Razorpay rail.

Dental AI Pipeline — Automated OPG Radiograph Analysis

PyTorch, YOLOv11, Docker

- Six-stage pipeline: tooth detection, FDI numbering, caries and periodontal bone-loss classification, structured reporting.
- Detection **mAP@0.5 0.810**; bone-loss **macro F1 0.9233**; **58.3%** exact FDI numbering measured end-to-end rather than per-stage.
- Externally validated on **four hospitals** plus a second public benchmark; 174 tests across 97.7M parameters; CPU-only Docker image.

SKILLS

AI / ML: Agentic AI systems, multi-agent orchestration, agent memory design, LLM prompting and tool use, context routing, Deep Learning (CNN, Transformer, RT-DETR), Computer Vision, Explainable AI (Grad-CAM, Integrated Gradients, Captum)

Languages: Python, Go, TypeScript, JavaScript, Java, Dart

Frameworks: PyTorch, scikit-learn, FastAPI, Next.js, React, Node.js, Spring Boot, Playwright, pandas, NumPy

Infrastructure: Docker, PostgreSQL, MongoDB, GitHub Actions CI/CD, SLURM, Vercel, Cloudflare R2, pytest

Practice: Experiment design (ablation, statistical significance), reproducibility and seed control, Scrum leadership, SOLID and GoF patterns

CERTIFICATIONS

Problem Solving (Basic) — HackerRank, Apr 2025 — hackerrank.com/certificates/236f2b956231